Supplementary Material

Gated input standardization and the search that led to it

S1. The technical improvement: gated input standardization (GIN)

The first convolution of YOLOv11n receives the image after a per-image, per-channel standardization that is blended with the identity by a learnable gate. For an input x (RGB, scaled to $[0, 1]$) with channel mean μ_c and standard deviation σ_c over the spatial positions,

$$\tilde{x}_c = x_c + a_c\Big(\frac{x_c - \mu_c}{\sqrt{\sigma_c^2 + \epsilon}}\, s + m - x_c\Big), \tag{1}$$

where a_c is a learnable gate per channel (initialized at 0.5), $m = 0.4$ and $s = 0.18$ are fixed constants that re-scale the standardized image to typical dataset statistics, and $\epsilon = 10^{-3}$. The module has three parameters (one gate per colour channel) and needs one mean and one variance per channel; its latency was not measured separately. It is active at training and inference time and is intended to remove the per-image colour cast and contrast differences that turbidity and backscatter introduce (red-channel loss, haze). The COCO weights of the remaining network are loaded unchanged. The released file `pwst_modules.py` contains the class `GatedIN` and the convolution wrapper `ConvGIN5`; the configuration is the YOLOv11n configuration with its first layer replaced by `ConvGIN5`. Training uses the protocol of Section IV-A of the manuscript with light augmentation (no mixup, no shear).

S2. All candidates screened by clip-grouped cross-validation

Four folds over 643 training and validation images, 100 epochs, no test data. Gains are mean differences from YOLOv11n in the same folds (clean mAP50 and the mean mAP50 over seven corruptions); the score is their average. The cross-validation of the later candidates (names starting with x_L, x_A, x_eca, x_se, x_p2) was not predictive of the full protocol (Section S3).

Code name	Candidate	Folds	Clean	Corrupted	Score	Positive
x_Agin5	SRAW-TM + GIN (blend 0.5)	4	+3.8	+4.0	+3.9	3/4
x_A_amp	SRAW-TM + AmpMix	4	+1.9	+3.8	+2.8	4/4
x_gin5	GIN, blend initialized at 0.5 (input only)	4	+3.1	+2.6	+2.8	3/4
x_Lgin5b	x_Lgin5 + GIN after the first stage	4	+2.7	+2.8	+2.8	3/4
x_amp	AmpMix (Fourier amplitude mixing, training only)	4	+1.6	+3.3	+2.4	3/4
x_Lgin5	x_L + GIN (blend 0.5)	4	+2.4	+2.4	+2.4	3/4
x_L_amp	x_L + AmpMix	4	+1.0	+2.8	+1.9	3/4
x_Agin5h8	SRAW-TM, 8 heads + GIN	3	+1.4	+2.0	+1.7	2/3
x_ms	MixStyle	4	+1.2	+2.1	+1.7	2/4
x_amplo	AmpMix, low frequencies	4	+1.1	+1.9	+1.5	3/4
x_ampp1	AmpMix on every sample	4	−0.6	+3.2	+1.3	3/4
x_L	SRAW-TM + depth-wise branch (ConvSRAL) at P4, P5	4	+1.2	+1.3	+1.2	3/4
x_Ar1	SRAW-TM, full attention at P5	4	+1.3	+0.9	+1.1	2/4
x_L7gin5	ConvSRAL at P5 only + GIN (blend 0.5)	4	+0.4	+1.4	+0.9	3/4
x_Lh8	x_L, 8 heads	4	+0.7	+0.8	+0.8	3/4
x_Lgin	x_L + GIN (blend 0)	4	+0.8	+0.7	+0.8	3/4
x_amphi	AmpMix, high frequencies	4	+0.4	+1.0	+0.7	1/4
x_Lr4	x_L, reduction 4	4	+0.5	+0.9	+0.7	2/4
x_L7	ConvSRAL at P5 only	3	−0.1	+1.3	+0.6	2/3
x_L3	ConvSRAL at P3, P4, P5	4	+0.8	+0.4	+0.6	3/4
x_eca	zero-gated channel attention (ECA) after stride-2 convolutions	4	−0.6	+1.6	+0.5	2/4
x_ampf	feature-level amplitude mixing	4	−0.6	+1.4	+0.4	3/4
x_Lse	x_L + SE	4	+0.7	−0.1	+0.3	2/4
x_Leca	x_L + ECA	3	−0.5	+0.8	+0.1	2/3
x_ush	unsharp masking	4	+0.1	0.0	0.0	2/4
x_Lush	x_L + unsharp masking	4	+0.3	−0.3	0.0	2/4
x_Llk	x_L + depth-wise branch in C3k2	4	−0.2	−0.4	−0.3	2/4
x_eca57	ECA at P4, P5	4	−1.3	0.0	−0.6	1/4
x_Lginf	x_L + instance normalization of early features	2	−1.7	+0.4	−0.7	1/2
pwA	SRAW-TM at P4, P5 (pw_A)	4	−1.5	−0.8	−1.2	2/4
x_lcn	local contrast normalization	4	−0.8	−1.7	−1.2	1/4
x_se	zero-gated squeeze-and-excitation	4	−2.0	−0.5	−1.3	1/4
x_Lr1	x_L, full attention	3	−0.8	−1.8	−1.3	1/3
x_Lsg	x_L + spectral gain	2	−3.2	−0.4	−1.8	0/2
x_p2	P2 detection head	3	−2.3	−2.7	−2.5	0/3
x_L5	ConvSRAL at P4 only	1	−2.7	−2.4	−2.5	0/1
x_p2gin	P2 head + GIN	2	−3.0	−3.1	−3.0	1/2

S3. Validation-split screen of round 7

Full protocol (300 epochs, light augmentation), five new seeds (70–74) per candidate, selection on the *validation* mAP50 only; baseline: YOLOv11n with light augmentation, validation mAP50 83.1 ± 0.6 (ten seeds). Rule fixed beforehand: gain of at least 0.7 points and at least four of five seeds above the baseline mean. The candidate that met the rule, P5-only SRAW-TM with depth-wise branch and GIN, did not keep its validation gain with new seeds (Table S4); x_L, SRAW-TM with depth-wise branch and GIN and SRAW-TM with GIN were stopped after two seeds because their test confirmations were already null.

Code name	Seeds	Validation mAP50	Gain	Seeds above baseline
x_L7gin5	5	84.21	+1.08	5/5
x_Lgin5b	4	83.61	+0.48	4/4
x_L7	5	83.51	+0.37	4/5
x_eca	5	83.50	+0.36	3/5
x_gin5	5	83.43	+0.29	4/5
x_Agin5h8	5	83.42	+0.28	4/5
x_Lh8	5	83.11	−0.02	3/5
x_Lse	5	83.11	−0.03	2/5
x_L	2	82.81	−0.32	1/2
x_Lr4	1	82.49	−0.64	0/1
x_Agin5	2	82.03	−1.11	0/2
x_Lgin5	2	81.64	−1.50	0/2

S4. All pre-registered confirmations on the test split

Mean ± standard deviation over the seeds in parentheses; differences from YOLOv11n with the full augmentation (eight seeds) or with light augmentation (ten seeds), as stated by the recipe of each row; Welch two-sided p values, uncorrected. "Named" are the nine corruptions of Section IV-F (blur, contrast, haze); "additional" are the twelve corruptions fixed before the AmpMix runs. The primary endpoints were: clean test mAP50 for all rows except GIN, whose primary endpoint (fixed in the queue file before the runs) was the mean mAP50 over the nine named corruptions. Of the primary endpoints listed here, only that of GIN meets its level.

Model (seeds)	n	Clean mAP50	Clean: difference	mAP50-95: difference	Named: difference	Additional: difference
SRAW-TM + AmpMix (seeds 10–19), light aug.	10	89.8 ± 1.1	-0.1 ($p = 0.80$)	$+0.2$ ($p = 0.52$)	$+1.8$ ($p < 0.001$)	$+4.5$ ($p < 0.001$)
x_L, full augmentation (30–39)	10	89.5 ± 0.7	$+0.3$ ($p = 0.55$)	$+0.3$ ($p = 0.42$)	$+0.6$ ($p = 0.13$)	$+1.2$ ($p = 0.12$)
x_L, light aug. (30–39)	10	89.9 ± 1.2	-0.1 ($p = 0.92$)	$+0.1$ ($p = 0.84$)	-0.4 ($p = 0.54$)	-0.3 ($p = 0.64$)
SRAW-TM with depth-wise branch and GIN, full augmentation (40–49)	10	89.5 ± 0.9	$+0.2$ ($p = 0.61$)	-0.3 ($p = 0.40$)	$+0.6$ ($p = 0.13$)	$+1.8$ ($p = 0.03$)
SRAW-TM with depth-wise branch and GIN, light aug. (40–49)	10	90.2 ± 1.0	$+0.3$ ($p = 0.51$)	$+0.8$ ($p = 0.06$)	$+1.6$ ($p < 0.001$)	0.0 ($p = 0.95$)
SRAW-TM with GIN, full augmentation (50–59)	10	89.5 ± 1.1	$+0.2$ ($p = 0.67$)	-0.1 ($p = 0.78$)	$+0.6$ ($p = 0.24$)	$+1.5$ ($p = 0.07$)
SRAW-TM with GIN, light aug. (50–59)	10	90.2 ± 1.3	$+0.3$ ($p = 0.58$)	$+0.6$ ($p = 0.28$)	$+0.6$ ($p = 0.20$)	-1.6 ($p = 0.03$)
P5-only SRAW-TM with depth-wise branch and GIN, full augmentation (80–89)	10	89.8 ± 0.9	$+0.5$ ($p = 0.28$)	-0.2 ($p = 0.54$)	$+0.8$ ($p = 0.10$)	$+1.6$ ($p = 0.05$)
P5-only SRAW-TM with depth-wise branch and GIN, light aug. (80–89)	10	90.3 ± 0.9	$+0.4$ ($p = 0.29$)	$+0.8$ ($p = 0.11$)	$+1.4$ ($p = 0.02$)	-0.7 ($p = 0.26$)
GIN, light aug. (90–99)	10	89.8 ± 0.9	-0.1 ($p = 0.87$)	$+0.3$ ($p = 0.59$)	$+1.3$ ($p = 0.008$)	-0.8 ($p = 0.26$)
YOLOv11n, TrashCan init., full augmentation (20–27)	8	90.5 ± 0.8	$+1.2$ ($p = 0.02$)	0.0 ($p = 1.00$)	$+0.9$ ($p = 0.03$)	-0.1 ($p = 0.89$)
YOLOv11n, TrashCan init., light aug. (20–27)	8	90.5 ± 1.0	$+0.6$ ($p = 0.18$)	$+0.6$ ($p = 0.15$)	$+0.2$ ($p = 0.67$)	-0.8 ($p = 0.08$)
SRAW-TM + AmpMix, TrashCan init. (20–27)	8	90.1 ± 1.0	$+0.2$ ($p = 0.62$)	$+0.3$ ($p = 0.44$)	$+2.1$ ($p < 0.001$)	$+4.9$ ($p < 0.001$)

S5. GIN under each condition

Mean mAP50 (± standard deviation) over ten seeds of GIN (seeds 90–99) and ten seeds of YOLOv11n, both with light augmentation; p values are uncorrected and the level of the primary endpoint applies to the mean over the nine named conditions only.

Condition	Group	YOLOv11n	GIN	Difference
clean	clean	89.8 ± 1.1	89.7 ± 1.0	-0.1 ($p = 0.91$)
blur_1	named	89.5 ± 0.8	89.6 ± 0.9	+0.1 ($p = 0.79$)
blur_2	named	80.1 ± 1.2	79.6 ± 2.9	-0.5 ($p = 0.65$)
blur_3	named	61.5 ± 2.7	60.5 ± 3.8	-1.0 ($p = 0.49$)
contrast_0.7	named	89.0 ± 1.0	89.8 ± 0.9	+0.7 ($p = 0.10$)
contrast_0.5	named	88.3 ± 0.9	89.2 ± 0.7	+0.9 ($p = 0.02$)
contrast_0.3	named	84.7 ± 1.0	87.0 ± 1.2	+2.2 ($p < 0.001$)
haze_0.8	named	89.0 ± 0.8	89.6 ± 0.9	+0.6 ($p = 0.12$)
haze_0.6	named	85.6 ± 1.1	87.6 ± 0.9	+2.0 ($p < 0.001$)
haze_0.4	named	76.6 ± 2.2	82.9 ± 1.3	+6.3 ($p < 0.001$)
motion_9	additional	77.9 ± 1.8	76.5 ± 3.0	-1.5 ($p = 0.21$)
motion_17	additional	51.7 ± 3.5	49.1 ± 4.7	-2.7 ($p = 0.17$)
motion_25	additional	37.9 ± 4.3	36.3 ± 4.6	-1.6 ($p = 0.42$)
defocus_2	additional	89.0 ± 1.0	89.1 ± 0.7	+0.1 ($p = 0.72$)
defocus_4	additional	77.6 ± 1.9	76.3 ± 2.9	-1.3 ($p = 0.24$)
defocus_6	additional	57.8 ± 3.4	56.0 ± 3.8	-1.8 ($p = 0.27$)
jpeg_30	additional	87.9 ± 0.8	86.7 ± 1.1	-1.2 ($p = 0.01$)
jpeg_15	additional	75.5 ± 1.3	72.0 ± 2.8	-3.4 ($p = 0.004$)
jpeg_8	additional	46.1 ± 1.7	41.6 ± 4.7	-4.5 ($p = 0.02$)
redloss_0.8	additional	87.2 ± 0.6	88.7 ± 1.0	+1.5 ($p = 0.001$)
redloss_0.6	additional	82.8 ± 1.4	85.8 ± 1.5	+3.0 ($p < 0.001$)
redloss_0.4	additional	78.5 ± 2.0	82.4 ± 2.0	+3.8 ($p < 0.001$)

S6. Initialization from TrashCan

A YOLOv11n checkpoint was trained for 100 epochs on TrashCan 1.0 (22 classes) after removing every frame of each video that contains an image with a thumbnail correlation of at least 0.95 with a mini-Trash validation or test image (3649 training and 781 validation images remain, 117 of 312 videos removed); the detection head was re-initialized. Eight new seeds (20–27) of YOLOv11n with the full augmentation, YOLOv11n with light augmentation and SRAW-TM + AmpMix with light augmentation were fine-tuned from it. None of the three comparisons fixed beforehand met the level of 0.0167 (Table S4, last three rows). Even after the exclusion, 17% of the test images have a retained TrashCan image with a thumbnail correlation of at least 0.90 (36% for the mini-Trash training images).

S7. Statistical details of statements in the manuscript

The manuscript states the conclusions of the tests in words. The table gives the test details that belong to each statement. All tests are two-sided Welch tests, and uncorrected means that no correction for multiple comparisons was applied.

No.	Statement in the manuscript ends with	Details
1	… with early PWAM is significantly lower than YOLOv11n	1.7 points in mAP50 on the validation split, $p = 0.01$
2	… at P3–P5, differs from YOLOv11n by +0.2 points	$p = 0.72$
3	… from PWAM-only at P3–P5 by - 0.5 points	$p = 0.68$
4	… points . Further placements of the same modules	SRAW-TM at P5, PWAM in the C3k2 blocks, PWAM in the neck, PWAM and SRA in the C3k2 blocks (three seeds each) and SRAW-TM at P3–P5 (one seed, test mAP50 90.4)
5	… from the amplitude-only variant by - 0.7 points	Welch $p = 0.65$
6	… and from the phase-only variant by +1.1 points	$p = 0.44$
7	… for PWST-YOLOv11, which is 8.6 points below YOLOv11n	Welch $p < 0.001$
8	… With light augmentation, YOLOv11n reaches 52.9±4.0 %	+0.5 points, $p = 0.75$
9	… augmentation alone, AmpMix changes it by +1.5 points	$p = 0.40$
10	… and SRAW-TM + AmpMix by - 0.5 points	$p = 0.77$

No.	Statement in the manuscript ends with	Details
11	… augmentation differs from YOLOv11n by - 0.9 points	$p = 0.41$
12	… corruptions rises from 71.4±1.0 to 74.2±1.6	+2.8 points, Welch $p = 0.01$
13	… augmentation, the test mAP50-95 rises by 3.6 points	Welch $p < 0.001$
14	… the twelve additional corruptions rises by 6.9 points	Welch $p < 0.001$
15	… and the test mAP50 changes by +0.8 points	$p = 0.12$
16	… including a confirmation with new seeds fixed beforehand	+0.5 points, $p = 0.34$

S8. All TrashCan runs

TrashCan 1.0 split 70%/20%/10% into training, validation and test images; 200 epochs, SGD, batch size 8, initial learning rate 0.01 with cosine decay, best checkpoint by validation fitness (Table I of the manuscript). Seeds in parentheses. Round 1 compared YOLOv11n, PWST-YOLOv11 and GIN with full augmentation. The three candidates of round 2 were fixed in `preregistrations.txt` before their runs, and the rule chose one of them on the validation mAP50 of seeds 0–2 against the YOLOv11n row of the same recipe; the chosen candidate, GIN with light augmentation, and its baseline were then trained with seeds 10–14 for the test reported in the manuscript. The first attempt of seed 1 of GIN with the gate initialized at 0 diverged (NaN losses in the first two warm-up epochs, validation mAP50 of 0 for 140 epochs) and was stopped; the row uses a second run of that seed. Differences are taken from the YOLOv11n row of the same recipe and seeds; "named" and "additional" are the mean mAP50 over the nine and the twelve corruptions of Section IV-F of the manuscript on the TrashCan test split. Welch two-sided p values, uncorrected.

	Model (seeds)	n	Val. mAP50	Val.: difference	Test mAP50	Test: difference	mAP50-95: difference	Named	Additional
Round 1	YOLOv11n, full augmentation (0–2)	3	64.0 ± 1.8	–	63.8 ± 1.5	–	–	51.7 ± 1.9	38.3 ± 1.9
	PWST-YOLOv11, full augmentation (0–2)	3	58.5 ± 7.7	-5.5 ($p = 0.34$)	61.2 ± 5.7	-2.6 ($p = 0.51$)	-1.8 ($p = 0.72$)	-2.7 ($p = 0.49$)	-2.9 ($p = 0.32$)
	GIN, full augmentation (0–2)	3	62.4 ± 0.8	-1.5 ($p = 0.28$)	59.8 ± 1.6	-4.0 ($p = 0.03$)	-2.2 ($p = 0.34$)	-4.1 ($p = 0.08$)	-5.1 ($p = 0.03$)
Round 2	GIN, gate initialized at 0, full augmentation (0–2)	3	63.7 ± 3.0	-0.2 ($p = 0.91$)	63.9 ± 3.5	+0.1 ($p = 0.95$)	+1.1 ($p = 0.71$)	–	–
	YOLOv11n, light aug. (0–2)	3	70.2 ± 2.8	–	71.5 ± 3.9	–	–	42.1 ± 4.7	46.2 ± 2.1
	GIN, light aug. (0–2)	3	73.2 ± 2.7	+3.0 ($p = 0.25$)	75.0 ± 2.6	+3.5 ($p = 0.27$)	+3.6 ($p = 0.17$)	+11.7 ($p = 0.03$)	+2.8 ($p = 0.19$)
Final test	YOLOv11n, light aug. (10–14)	5	72.6 ± 4.5	–	74.1 ± 3.7	–	–	45.1 ± 3.7	47.5 ± 2.9
	GIN, light aug. (10–14)	5	75.3 ± 3.3	+2.7 ($p = 0.32$)	76.8 ± 4.1	+2.7 ($p = 0.30$)	+2.8 ($p = 0.19$)	+9.3 ($p = 0.002$)	+2.5 ($p = 0.22$)

S9. The final TrashCan pair under each corruption

Mean mAP50 (± standard deviation) on the TrashCan test split over seeds 10–14 of YOLOv11n and of GIN, both with light augmentation; p values are uncorrected.

Condition	Group	YOLOv11n	GIN	Difference
clean	clean	74.2 ± 3.7	76.7 ± 3.9	+2.6 ($p = 0.32$)
blur_1	named	67.0 ± 2.9	71.0 ± 2.5	+4.0 ($p = 0.05$)
blur_2	named	43.6 ± 3.3	44.3 ± 1.4	+0.8 ($p = 0.66$)
blur_3	named	28.6 ± 2.9	26.3 ± 1.3	-2.3 ($p = 0.16$)
contrast_0.7	named	70.7 ± 5.4	76.0 ± 3.5	+5.3 ($p = 0.11$)
contrast_0.5	named	59.7 ± 6.6	71.6 ± 3.5	+11.9 ($p = 0.01$)
contrast_0.3	named	30.7 ± 4.7	50.6 ± 3.0	+19.9 ($p < 0.001$)
haze_0.8	named	63.0 ± 6.2	72.4 ± 5.0	+9.3 ($p = 0.03$)
haze_0.6	named	31.4 ± 3.1	53.3 ± 6.2	+21.9 ($p < 0.001$)
haze_0.4	named	11.1 ± 1.8	23.8 ± 3.9	+12.7 ($p < 0.001$)
motion_9	additional	47.2 ± 1.8	48.8 ± 1.8	+1.6 ($p = 0.20$)
motion_17	additional	21.3 ± 2.9	22.1 ± 0.7	+0.8 ($p = 0.58$)
motion_25	additional	13.9 ± 2.3	14.9 ± 1.3	+1.0 ($p = 0.42$)
defocus_2	additional	64.3 ± 2.8	69.0 ± 2.7	+4.7 ($p = 0.03$)
defocus_4	additional	40.8 ± 3.0	40.5 ± 1.2	-0.3 ($p = 0.85$)
defocus_6	additional	24.8 ± 2.1	22.3 ± 1.6	-2.5 ($p = 0.07$)
jpeg_30	additional	72.4 ± 3.2	75.1 ± 3.9	+2.7 ($p = 0.27$)
jpeg_15	additional	67.6 ± 3.9	68.6 ± 4.9	+1.0 ($p = 0.73$)
jpeg_8	additional	49.3 ± 3.2	48.5 ± 6.3	-0.8 ($p = 0.80$)
redloss_0.8	additional	63.6 ± 5.8	70.9 ± 6.6	+7.3 ($p = 0.10$)
redloss_0.6	additional	54.4 ± 5.8	62.3 ± 8.3	+7.9 ($p = 0.12$)
redloss_0.4	additional	50.5 ± 6.1	57.0 ± 7.8	+6.6 ($p = 0.18$)